

Zubair Mughal

LIVE CODING CLASSES FOR KIDS, TEENS & BEGINNERS

SQL

COURSE 10 OF 12
CURRICULUM GUIDE

SQL for Teens & Beginners

How real applications store & query data

GRADE BAND

Grade 9–12

DIFFICULTY

Beginner

DURATION

8 weeks · 2 sessions/week

ABOUT THIS COURSE

Course Overview

Almost every real application is backed by a database. Students learn to design relational schemas and write SQL queries to create, read, update and analyze structured data.

GRADE BAND Grade 9–12	DIFFICULTY Beginner	DURATION 8 weeks · 2 sessions/week
---	---	--

Prerequisites: None.

LEARNING OUTCOMES

What Students Will Learn

- Understand relational database concepts: tables, rows and keys
- Write SELECT queries with filtering, sorting and aggregation
- Join data across multiple related tables
- Insert, update and delete data safely
- Design a simple, normalized database schema
- Apply SQL to answer real analytical questions

WHAT WE'LL USE

Tools & Technologies

SQL (SQLite / PostgreSQL)	DB Browser for SQLite	VS Code
---------------------------	-----------------------	---------

CURRICULUM

Module-by-Module Breakdown

MODULE	FOCUS
01 · Databases & Tables	Relational concepts and keys.
02 · SELECT & Filtering	WHERE, ORDER BY, LIMIT.
03 · Aggregation	COUNT, SUM, AVG, GROUP BY.

MODULE	FOCUS
04 · Joins	Combining data across related tables.
05 · Modifying Data	INSERT, UPDATE, DELETE — safely.
06 · Database Design	Designing a simple schema.
07 · Real-World Queries	Solving analytical mini-challenges.
08 · Capstone Database Project	Build and query a small database.

LEARNING BY BUILDING

Projects Students Will Build

- Movie / Music Library Database — Schema design and querying.
- School / Club Management Database — Multi-table relationships.
- Capstone Database + Query Set — A designed database with real analytical queries.

BEYOND THE SYNTAX

Skills Developed

Relational thinking SQL querying Database design Data analysis fundamentals

PORTFOLIO OUTCOME

A working database schema with a documented set of queries answering real questions.

RECOMMENDED NEXT STEP

Node.js for Teens & Beginners, or Data Science for Teens